

WEEK 3 – ADVANCED CYBERSECURITY PRACTICAL TASK

TASK 7 – Network Discovery & Basic Nmap Scanning

Objective

Understand how a security analyst discovers devices and exposed network services.

Tools Used

- Kali Linux
- Metasploitable
- VMware Workstation/VirtualBox
- Nmap

Practical Work

First, the network configuration of the laboratory environment was identified using the `ip addr` and `ip route` commands.

Kali Linux –

IP Address - 192.168.27.128

Subnet – 255.255.255.0 (/24)

Default Gateway - 192.168.27.2

Lab network - 192.168.27.0/24

Metasploitable 2-

IP Address - 192.168.27.129

Subnet – 255.255.255.0

Default Gateway - 192.168.27.2

Host Discovery

Nmap host discovery was performed using: `nmap -sn [Lab-Network]/24`

```
└─$ nmap -sn 192.168.27.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-17 10:29 -0500
Nmap scan report for 192.168.27.1
Host is up (0.00037s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.27.2
Host is up (0.00038s latency).
MAC Address: 00:50:56:F3:49:2C (VMware)
Nmap scan report for 192.168.27.254
Host is up (0.00085s latency).
MAC Address: 00:50:56:F6:93:73 (VMware)
Nmap scan report for 192.168.27.128
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.41 seconds
```

Basic Nmap Scan

A basic Nmap scan was performed against the authorized target VM using:
nmap (target IP)

```
└─$ nmap 192.168.27.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-18 09:16 -0500
Nmap scan report for 192.168.27.129
Host is up (0.0070s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:FA:DD:2A (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.14 seconds
```

Observations

The host discovery scan successfully identified active devices within the authorized laboratory network. The basic Nmap scan then identified open ports and the services associated with those ports.

🔴 TASK 8 – Advanced Nmap Security Assessment

Objective

Perform a more detailed assessment of your authorized lab machines.

Kali Linux → Scanner

Metasploitable2 → Target

Required scans:

Service/version detection: nmap -sV 192.168.27.129

1. Nmap results

```
(nehawalmiki@kali) - [~/Desktop]
$ nmap -sV 192.168.27.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 10:14 -0500
Nmap scan report for 192.168.27.129
Host is up (0.0027s latency).
Not shown: 978 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8180/tcp  open  unknown
MAC Address: 00:0C:29:FA:DD:2A (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 191.92 seconds
```

2. Results Table

IP Address	Port	Protocol	Service	Version	Risk	Observation
192.168.27.129	21	TCP	FTP	vsftpd 2.3.4	High	FTP service exposed for file transfer
192.168.27.129	22	TCP	SSH	OpenSSH 4.7p1	Medium	Remote administration service exposed
192.168.27.129	23	TCP	Telnet	Linux telnetd	High	Remote login service using an insecure plaintext protocol
192.168.27.129	80	TCP	HTTP	Apache 2.2.8	High	Web server exposed
192.168.27.129	1099	TCP	JAVA RMI	GNU Classpath rmiregistry	High	Java remote registry exposed
192.168.27.129	1524	TCP	Bindshell	Metasploitable root shell	Critical	Root shell service exposed in the lab

192.168.27.129	2049	TCP	NFS	RPC #100003	High	Network File System service exposed
192.168.27.129	3306	TCP	MySQL	5.0.51a-3ubuntu5	High	Database service exposed
192.168.27.129	5432	TCP	PostgreSQL	8.3.0–8.3.7	High	Database service exposed
192.168.27.129	5900	TCP	VNC	Protocol 3.3	High	Remote graphical access service exposed

3. Advanced Analysis

Port 21 – FTP

1. What service is running?

FTP is running using vsftpd 2.3.4.

2. Why is the service required?

FTP can be used for transferring files between the client and server.

3. Is it necessary to expose it?

Only if file-transfer functionality is actually required.

4. What could happen if the service is vulnerable?

An attacker could potentially gain unauthorized access, retrieve files, upload malicious files, or exploit vulnerabilities in the FTP server.

5. How could it be secured?

- Upgrade to a supported version.
- Disable anonymous access if unnecessary.

Port 22 – SSH

Service: OpenSSH 4.7p1

SSH provides secure remote administration.

Why required: Administrators can remotely manage the Linux system.

Exposure: It should only be accessible from trusted management networks where possible.

Potential impact: A vulnerable or poorly configured SSH service could allow unauthorized remote access.

Security measures:

- Upgrade the SSH server.
- Use strong authentication.

Port 23 – Telnet

Service: Linux telnetd

Telnet provides remote command-line access but does not provide the same encryption protections as SSH.

Risk: High.

Potential impact: Credentials and other communication can potentially be exposed to network interception.

Security measures:

- Disable Telnet.
- Use SSH instead.

Port 80 – HTTP

Service: Apache httpd 2.2.8

HTTP provides web-server functionality.

Why required: It may host websites or web applications.

Potential impact: Vulnerabilities in the web server or hosted application could potentially result in unauthorized access, information disclosure, or compromise.

Security measures:

- Upgrade to a supported Apache version.
- Remove unnecessary web applications.

Port 1524 – Bindshell

This is particularly important in your lab because Nmap identifies:

1524/tcp open bindshell Metasploitable root shell

This is an intentionally insecure service associated with the Metasploitable training environment.

Why required?

It isn't normally required on a secure production system. It exists as part of the vulnerable training environment.

Potential impact:

A remotely accessible root shell represents extremely serious system compromise if present on a real system.

Security measures:

- Remove/disable the service.
- Restrict the port using firewall rules.

Port 3306 – MySQL

Service: MySQL 5.0.51a

Why required: Database applications may require MySQL to store and retrieve data.

Potential impact: An exposed or vulnerable database could result in unauthorized access to application data.

Security measures:

- Upgrade to a supported version.
- Restrict database access to application servers.

Port 5432 – PostgreSQL

Service: PostgreSQL 8.3.x

The database service may be required by applications that use PostgreSQL.

Potential impact: Unauthorized database access could expose, modify, or delete stored information.

Security measures:

- Upgrade to a supported PostgreSQL release.
- Restrict network access.

Port 5900 – VNC

Service: VNC protocol 3.3

VNC provides remote graphical access.

Why required: Administrators may use it for remote GUI access.

Potential impact: If poorly protected, unauthorized users could potentially obtain remote access to the graphical desktop.

Security measures:

- Restrict VNC access through a firewall.
- Use secure tunneling such as SSH where appropriate.

OS detection: `sudo nmap -O 192.168.27.129`

```
(nehawalmiki@kali)-[~/Desktop]
$ sudo nmap -O 192.168.27.129
[sudo] password for nehawalmiki:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 11:09 -0500
Nmap scan report for 192.168.27.129
Host is up (0.0020s latency).
Not shown: 978 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8180/tcp  open  unknown
MAC Address: 00:0C:29:FA:DD:2A (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.20 seconds
```

Nmap OS detection identified the target system as a general-purpose Linux machine. The detected operating system is within the Linux 2.6.x family, with the scan estimating Linux kernel versions between 2.6.9 and 2.6.33. The target was one network hop away from the scanning machine. The system exposed multiple TCP services, indicating a broad network attack surface.

TCP SYN scan: `sudo nmap -sS 192.168.27.129`

```
(nehawalmiki@kali)-[~/Desktop]
└─$ sudo nmap -sS 192.168.27.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 11:12 -0500
Nmap scan report for 192.168.27.129
Host is up (0.0014s latency).
Not shown: 978 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8180/tcp  open  unknown
MAC Address: 00:0C:29:FA:DD:2A (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.32 seconds
```

A TCP SYN scan was performed against the authorized Metasploitable laboratory machine at 192.168.27.129. The scan identified 22 open TCP ports and 978 closed TCP ports. The results were consistent with the service/version detection scan, confirming that multiple TCP services are accessible on the target system. The large number of exposed services indicates a broad attack surface, which is expected in the intentionally vulnerable Metasploitable environment.

Limited UDP scan: `sudo nmap -sU --top-ports 20 192.168.27.129`

```
[sudo] password for nehawalmiki:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 12:56 -0500
Nmap scan report for 192.168.27.129
Host is up (0.0019s latency).

PORT      STATE      SERVICE
53/udp    open       domain
67/udp    open|filtered dhcps
68/udp    open|filtered dhcpc
69/udp    open|filtered tftp
123/udp   closed     ntp
135/udp   closed     msrpc
137/udp    open       netbios-ns
138/udp    open|filtered netbios-dgm
139/udp    open|filtered netbios-ssn
161/udp    closed     snmp
162/udp    closed     snmptrap
445/udp    open|filtered microsoft-ds
500/udp    closed     isakmp
514/udp    closed     syslog
520/udp    closed     route
631/udp    open|filtered ipp
1434/udp   open|filtered ms-sql-m
1900/udp   closed     upnp
4500/udp   closed     nat-t-ike
49152/udp  open|filtered unknown
MAC Address: 00:0C:29:FA:DD:2A (VMware)

Nmap done: 1 IP address (1 host up) scanned in 7.01 seconds
```

A limited UDP scan of the top 20 UDP ports was performed against the authorized lab machine 192.168.27.129. The scan identified UDP port **53 (DNS)** and UDP port **137 (NetBIOS Name Service)** as open. Several other ports were reported as **open|filtered**, meaning Nmap could not definitively determine whether they were open because UDP responses may be absent or filtered by a firewall.

Vulnerability NSE scripts: `nmap --script vuln 192.168.27.129`

```
(nehawalmiki@kali) - [~/Desktop]
$ nmap --script vuln 192.168.27.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-19 13:24 -0500
Nmap scan report for 192.168.27.129
Host is up (0.0024s latency).
Not shown: 978 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|     vsFTPD version 2.3.4 backdoor
|       State: VULNERABLE (Exploitable)
|       IDs: BID:48539 CVE:CVE-2011-2523
|       vsFTPD version 2.3.4 backdoor, this was reported on 2011-07-04.
|       Disclosure date: 2011-07-03
|       Exploit results:
|         Shell command: id
|         Results: uid=0(root) gid=0(root)
|       References:
|         http://scarybeastsecurity.blogspot.com/2011/07/alert-vsftpd-download-backdoored.html
|         https://www.securityfocus.com/bid/48539
|         https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|         https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
| smtp-vuln-cve2010-4344:
|   The SMTP server is not Exim: NOT VULNERABLE
| ssl-dh-params:
|   VULNERABLE:
|     Anonymous Diffie-Hellman Key Exchange MitM Vulnerability
|       State: VULNERABLE
|       Transport Layer Security (TLS) services that use anonymous
|       Diffie-Hellman key exchange only provide protection against passive
|       eavesdropping, and are vulnerable to active man-in-the-middle attacks
|       which could completely compromise the confidentiality and integrity
|       of any data exchanged over the resulting session.
|       Check results:
|         ANONYMOUS DH GROUP 1
|         Cipher Suite: TLS_DH_anon_WITH_DES_CBC_SHA
```

Port	Service	Vulnerability	Risk	Observation
21/TCP	FTP – vsftpd 2.3.4	vsftpd 2.3.4 backdoor (CVE-2011-2523)	Critical	NSE identified a known backdoor vulnerability; the lab result indicates root-level shell access

NSE vulnerability scanning identified significant security issues in the authorized Metasploitable lab system, including the vsftpd 2.3.4 backdoor vulnerability on port 21 and an anonymous Diffie-Hellman TLS configuration on the SMTP service. These findings demonstrate the risks associated with outdated software and insecure cryptographic configurations.

TASK 9 – Wireshark Network Traffic Capture

Wireshark Filters Used

- dns
- tcp
- udp
- icmp
- arp
- tcp.flags.syn == 1
- tcp.flags.reset == 1
- dns.flags.response == 0

Packet Analysis

The captured traffic was analyzed for DNS, TCP, UDP and ARP packets.

tcp.flags.syn == 1

No.	Time	Source	Destination	Protocol	Length	Info
462	27.892618500	192.168.0.102	142.250.146.95	TCP	66	18073 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
466	27.900512100	142.250.146.95	192.168.0.102	TCP	66	443 → 18073 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
480	27.922858700	192.168.0.102	104.16.79.73	TCP	66	34939 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
585	27.991242900	104.16.79.73	192.168.0.102	TCP	66	443 → 34939 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 SACK_PERM WS=8192
1422	28.507030900	192.168.0.102	172.217.119.4	TCP	66	12434 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1440	28.512945300	172.217.119.4	192.168.0.102	TCP	66	443 → 12434 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
1698	28.638580300	192.168.0.102	163.181.214.12	TCP	66	41939 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1926	28.765483300	163.181.214.12	192.168.0.102	TCP	66	443 → 41939 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1440 SACK_PERM WS=512
2156	30.613925700	192.168.0.102	139.45.197.106	TCP	66	53406 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
2159	30.748884300	139.45.197.106	192.168.0.102	TCP	58	443 → 53406 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1440
2200	31.092299700	192.168.0.102	23.52.73.105	TCP	66	53407 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
2201	31.135120400	23.52.73.105	192.168.0.102	TCP	66	443 → 53407 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1384 SACK_PERM WS=128
2202	31.135121500	23.52.73.105	192.168.0.102	TCP	66	[TCP Retransmission] 443 → 53407 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1384 SACK_PERM WS=128

A TCP SYN-ACK retransmission was observed. This indicates that the SYN-ACK packet was retransmitted, which can occur when the expected acknowledgment is not received within the required time. A single retransmission does not by itself indicate malicious activity.

tcp.flags.reset == 1

tcp.flags.reset == 1					
No.	Time	Source	Destination	Protocol	Length Info
348	27.742846800	150.171.27.10	192.168.0.102	TCP	54 443 → 44848 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2313	32.421184400	150.171.27.12	192.168.0.102	TCP	54 443 → 8235 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2346	35.440446600	150.171.27.10	192.168.0.102	TCP	54 443 → 56675 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

dns.flags.response == 0

dns.flags.response == 0					
No.	Time	Source	Destination	Protocol	Length Info
63	24.910394600	192.168.0.102	192.168.0.1	DNS	74 Standard query 0xca64 HTTPS www.google.com
64	24.911820100	192.168.0.102	192.168.0.1	DNS	74 Standard query 0xd849 A www.google.com
77	26.084354700	192.168.0.102	192.168.0.1	DNS	71 Standard query 0x6420 HTTPS chatgpt.com
78	26.084970900	192.168.0.102	192.168.0.1	DNS	71 Standard query 0xd9d1 A chatgpt.com
255	27.450783100	192.168.0.102	192.168.0.1	DNS	70 Standard query 0x8c15 HTTPS anikoto.cz
256	27.451541600	192.168.0.102	192.168.0.1	DNS	70 Standard query 0x21d7 A anikoto.cz
305	27.607171200	192.168.0.102	192.168.0.1	DNS	88 Standard query 0x053a HTTPS ogads-pa.clients6.google.com
306	27.607606400	192.168.0.102	192.168.0.1	DNS	88 Standard query 0xe2c9 A ogads-pa.clients6.google.com
353	27.788930300	192.168.0.102	192.168.0.1	DNS	80 Standard query 0xace2 HTTPS cdnjs.cloudflare.com
354	27.789531500	192.168.0.102	192.168.0.1	DNS	80 Standard query 0x3c4a A cdnjs.cloudflare.com
379	27.828722800	192.168.0.102	192.168.0.1	DNS	86 Standard query 0x6f85 HTTPS encrypted-tbn0.gstatic.com
380	27.829228700	192.168.0.102	192.168.0.1	DNS	86 Standard query 0x9896 A encrypted-tbn0.gstatic.com
455	27.884536300	192.168.0.102	192.168.0.1	DNS	80 Standard query 0x0e3c HTTPS fonts.googleapis.com
456	27.885681600	192.168.0.102	192.168.0.1	DNS	80 Standard query 0x4c35 A fonts.googleapis.com
471	27.906348800	192.168.0.102	192.168.0.1	DNS	89 Standard query 0x7951 HTTPS static.cloudflareinsights.com
472	27.906861500	192.168.0.102	192.168.0.1	DNS	89 Standard query 0x3b65 A static.cloudflareinsights.com
725	28.097108900	192.168.0.102	192.168.0.1	DNS	75 Standard query 0x9be3 HTTPS nekostream.site
726	28.098876400	192.168.0.102	192.168.0.1	DNS	75 Standard query 0x95e2 A nekostream.site
745	28.196247800	192.168.0.102	192.168.0.1	DNS	75 Standard query 0x1204 HTTPS play.google.com

DNS

dns					
No.	Time	Source	Destination	Protocol	Length Info
1267	28.430587000	192.168.0.102	192.168.0.1	DNS	91 Standard query 0x25a7 A content-autofill.googleapis.com
1358	28.477082900	192.168.0.102	192.168.0.1	DNS	78 Standard query 0x56b3 HTTPS dump.mewcdn.online
1359	28.477781900	192.168.0.102	192.168.0.1	DNS	78 Standard query 0x741e A dump.mewcdn.online
1407	28.501590700	192.168.0.1	192.168.0.102	DNS	479 Standard query response 0xfcc8 HTTPS fonts.gstatic.com A 192.178.193.94 NS h.root-servers.net NS f.root-servers.net NS a.root-servers.net NS i.root-servers.net
1408	28.501692800	192.168.0.1	192.168.0.102	DNS	470 Standard query response 0xf4d0 A fonts.gstatic.com A 192.178.193.94 NS h.root-servers.net NS e.root-servers.net NS l.root-servers.net NS o.root-servers.net
1410	28.502632000	192.168.0.1	192.168.0.102	DNS	148 Standard query response 0x0d67 HTTPS content-autofill.googleapis.com SOA ns1.google.com
1411	28.502783700	192.168.0.1	192.168.0.102	DNS	540 Standard query response 0x25a7 A content-autofill.googleapis.com A 172.217.119.4 A 172.217.117.4 A 172.217.112.4 A 172.217.118.4
1541	28.560164000	192.168.0.1	192.168.0.102	DNS	149 Standard query response 0x56b3 HTTPS dump.mewcdn.online SOA darling.ns.cloudflare.com
1674	28.633959200	192.168.0.1	192.168.0.102	DNS	471 Standard query response 0x741e A dump.mewcdn.online A 163.181.214.12 NS h.root-servers.net NS a.root-servers.net NS j.root-servers.net NS m.root-servers.net
1937	28.855118700	192.168.0.102	192.168.0.1	DNS	87 Standard query 0x4404 HTTPS safebrowsing.googleapis.com
1938	28.856121200	192.168.0.102	192.168.0.1	DNS	87 Standard query 0x9eed A safebrowsing.googleapis.com
1958	28.868021600	192.168.0.1	192.168.0.102	DNS	144 Standard query response 0x4404 HTTPS safebrowsing.googleapis.com SOA ns1.google.com
1974	28.879183400	192.168.0.1	192.168.0.102	DNS	480 Standard query response 0x9eed A safebrowsing.googleapis.com A 192.178.174.95 NS l.root-servers.net NS g.root-servers.net NS m.root-servers.net NS o.root-servers.net
2037	29.805312500	192.168.0.102	192.168.0.1	DNS	69 Standard query 0xe7f5 HTTPS llvpn.com
2038	29.805637200	192.168.0.102	192.168.0.1	DNS	69 Standard query 0x4224 A llvpn.com
2039	29.810396500	192.168.0.1	192.168.0.102	DNS	530 Standard query response 0xe7f5 HTTPS llvpn.com HTTPS NS c.root-servers.net NS g.root-servers.net NS m.root-servers.net NS f.root-servers.net
2040	29.810400100	192.168.0.1	192.168.0.102	DNS	478 Standard query response 0x4224 A llvpn.com A 172.67.154.229 A 184.21.13.59 NS a.root-servers.net NS m.root-servers.net NS j.root-servers.net NS o.root-servers.net
2120	29.924494200	192.168.0.102	192.168.0.1	DNS	73 Standard query 0x8b46 HTTPS my.rtmk.net
2121	29.925003500	192.168.0.102	192.168.0.1	DNS	73 Standard query 0xd514 A my.rtmk.net

ARP

arp					
No.	Time	Source	Destination	Protocol	Length Info
43	13.863273300	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
44	15.874140900	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
45	16.898404600	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
46	17.922328600	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
52	18.946166700	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
54	19.822042000	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
55	20.810935700	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
56	21.813539600	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
61	23.861658100	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
62	24.889429700	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
349	27.764942800	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
1905	28.753284000	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
2031	29.746546200	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
2158	30.742395400	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
2224	31.741000500	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
2314	32.723804000	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
2315	33.721966800	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
2345	34.715733800	TPLink_d5:d2:7c	Broadcast	ARP	42 Who has 192.168.0.106? Tell 192.168.0.1
2347	35.475110000	TPLink_d5:d2:7c	HonHaiPrecis 3e:fa:...	ARP	42 Who has 192.168.0.102? Tell 192.168.0.1

UDP

No.	Time	Source	Destination	Protocol	Length	Info
18	6.490009300	192.168.1.1	239.255.255.250	SSDP	387	NOTIFY * HTTP/1.1
19	6.493177600	192.168.1.1	239.255.255.250	SSDP	316	NOTIFY * HTTP/1.1
20	6.496818600	192.168.1.1	239.255.255.250	SSDP	375	NOTIFY * HTTP/1.1
21	6.500492000	192.168.1.1	239.255.255.250	SSDP	369	NOTIFY * HTTP/1.1
47	18.654965900	192.168.0.102	142.251.152.119	UDP	71	53106 → 443 Len=29
48	18.732914200	192.168.0.102	142.251.152.119	UDP	71	53106 → 443 Len=29
49	18.748092800	142.251.152.119	192.168.0.102	UDP	68	443 → 53106 Len=26
50	18.842422000	192.168.0.102	142.251.152.119	UDP	71	53106 → 443 Len=29
51	18.844255100	142.251.152.119	192.168.0.102	UDP	69	443 → 53106 Len=27
53	18.950759700	142.251.152.119	192.168.0.102	UDP	71	443 → 53106 Len=29
63	24.910394600	192.168.0.102	192.168.0.1	DNS	74	Standard query 0xca64 HTTPS www.google.com
64	24.911820100	192.168.0.102	192.168.0.1	DNS	74	Standard query 0xd849 A www.google.com
68	24.989213200	192.168.0.1	192.168.0.102	DNS	263	Standard query response 0xca64 HTTPS www.google.com HTTPS NS ns1.google.com NS ns2.google.com NS ns3.google.com NS ns4.google.com
69	24.989214300	192.168.0.1	192.168.0.102	DNS	366	Standard query response 0xd849 A www.google.com A 142.251.152.119 A 142.251.156.119 A 142.251.155.119 A 142.251.154.119 A 142.251.153.119
70	24.990579100	192.168.0.102	142.251.152.119	UDP	71	53106 → 443 Len=29
71	24.999166400	142.251.152.119	192.168.0.102	UDP	71	443 → 53106 Len=29
75	26.003094300	192.168.0.102	142.251.152.119	UDP	71	53106 → 443 Len=29
76	26.003528200	192.168.0.102	142.251.152.119	UDP	71	53106 → 443 Len=29
77	26.004354700	192.168.0.102	192.168.0.1	DNS	71	Standard query 0x6420 HTTPS chatgpt.com

TCP

No.	Time	Source	Destination	Protocol	Length	Info
26	7.991510600	23.52.73.19	192.168.0.102	TCP	66	443 → 7812 [ACK] Seq=1 Ack=2 Win=589 Len=0 SLE=1 SRE=2
28	9.629400600	104.18.39.21	192.168.0.102	TLSv1.2	78	Application Data
29	9.630435600	192.168.0.102	104.18.39.21	TLSv1.2	82	Application Data
30	9.732321500	104.18.39.21	192.168.0.102	TCP	54	443 → 54437 [ACK] Seq=25 Ack=29 Win=19 Len=0
33	11.510343500	192.168.0.102	52.168.117.170	TCP	55	43940 → 443 [ACK] Seq=1 Ack=1 Win=250 Len=1
34	11.780054100	52.168.117.170	192.168.0.102	TCP	66	443 → 43940 [ACK] Seq=1 Ack=2 Win=16385 Len=0 SLE=1 SRE=2
36	11.994670500	192.168.0.102	23.223.244.35	TCP	55	8133 → 443 [ACK] Seq=1 Ack=1 Win=253 Len=1
37	12.086249100	23.223.244.35	192.168.0.102	TCP	66	443 → 8133 [ACK] Seq=1 Ack=2 Win=1834 Len=0 SLE=1 SRE=2
40	12.507838100	192.168.0.102	23.215.4.24	TCP	55	1707 → 443 [ACK] Seq=1 Ack=1 Win=250 Len=1
41	12.523937700	23.215.4.24	192.168.0.102	TCP	66	443 → 1707 [ACK] Seq=1 Ack=2 Win=590 Len=0 SLE=1 SRE=2
57	23.089273000	192.168.0.102	163.70.144.60	TCP	124	57945 → 5222 [PSH, ACK] Seq=71 Ack=73 Win=252 Len=70 [TCP PDU reassembled in 57]
58	23.147071800	163.70.144.60	192.168.0.102	TCP	54	5222 → 57945 [ACK] Seq=73 Ack=141 Win=1986 Len=0
59	23.349960700	163.70.144.60	192.168.0.102	TCP	126	5222 → 57945 [PSH, ACK] Seq=73 Ack=141 Win=1986 Len=72 [TCP PDU reassembled in 59]
60	23.401309500	192.168.0.102	163.70.144.60	TCP	54	57945 → 5222 [ACK] Seq=141 Ack=145 Win=251 Len=0
65	24.915660300	192.168.0.102	104.18.39.21	TLSv1.2	144	Application Data
66	24.988303300	104.18.39.21	192.168.0.102	TCP	54	443 → 54437 [ACK] Seq=25 Ack=119 Win=19 Len=0
67	24.988307600	104.18.39.21	192.168.0.102	TLSv1.2	138	Application Data
72	25.037096400	192.168.0.102	104.18.39.21	TCP	54	54437 → 443 [ACK] Seq=119 Ack=109 Win=253 Len=0
73	26.042411500	192.168.0.102	104.18.39.21	TLSv1.2	144	Application Data

TASK 10 – Nmap + Wireshark Investigation

nmap -sS 192.168.81.129

```
(nehawalmiki@kali) - [~/Desktop]
$ nmap -sS 192.168.27.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-21 11:07 -0500
Nmap scan report for 192.168.27.129
Host is up (0.0014s latency).
Not shown: 978 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:FA:DD:2A (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.17 seconds
```

tcp.flags.reset == 1					
No.	Time	Source	Destination	Protocol	Length Info
138	17.463489400	23.212.254.42	192.168.0.102	TCP	54 443 → 40927 [RST] Seq=1 Win=0 Len=0
tcp.flags.syn == 1 && tcp.flags.ack == 1					
No.	Time	Source	Destination	Protocol	Length Info
12	0.448959400	142.250.134.94	192.168.0.102	TCP	66 80 → 46044 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
217	22.034012900	52.123.128.14	192.168.0.102	TCP	66 443 → 46045 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
314	30.977388000	23.212.254.107	192.168.0.102	TCP	66 443 → 41998 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1384 SACK_PERM WS=128
tcp.flags.syn == 1					
No.	Time	Source	Destination	Protocol	Length Info
11	0.363096000	192.168.0.102	142.250.134.94	TCP	66 46044 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
12	0.448959400	142.250.134.94	192.168.0.102	TCP	66 80 → 46044 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
216	22.008358100	192.168.0.102	52.123.128.14	TCP	66 46045 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
217	22.034012900	52.123.128.14	192.168.0.102	TCP	66 443 → 46045 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1440 WS=256 SACK_PERM
312	30.953252200	192.168.0.102	23.212.254.107	TCP	66 41998 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
314	30.977388000	23.212.254.107	192.168.0.102	TCP	66 443 → 41998 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1384 SACK_PERM WS=128

Nmap Result / Observed Port	Wireshark Evidence	Packet No.	Interpretation
Port 80 – reachable/open	192.168.0.102 → 142.250.134.94:80 [SYN], followed by 142.250.134.94:80 → 192.168.0.102 [SYN, ACK]	11, 12	SYN/ACK response indicates that TCP port 80 is reachable and accepting connections.
Port 443 – reachable/open	192.168.0.102 → 52.123.128.14:443 [SYN], followed by 52.123.128.14:443 → 192.168.0.102 [SYN, ACK]	216, 217	216, 217

TASK 11 – Advanced Network Traffic Investigation

1. Top Communicating Hosts

IPv4 · 18									
Address	Packets	Bytes	Total Packets	Percent Filtered	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Count
23.52.73.19	2	121 bytes	2	100.00%	1	66 bytes	1	55 bytes	1
23.215.4.24	2	121 bytes	2	100.00%	1	66 bytes	1	55 bytes	1
23.223.244.35	2	121 bytes	2	100.00%	1	66 bytes	1	55 bytes	1
52.168.117.170	2	121 bytes	2	100.00%	1	66 bytes	1	55 bytes	1
150.171.27.12	1	54 bytes	1	100.00%	1	54 bytes	0	0 bytes	1
192.178.211.188	2	121 bytes	2	100.00%	1	66 bytes	1	55 bytes	1
150.171.27.10	2	108 bytes	2	100.00%	2	108 bytes	0	0 bytes	1
104.26.7.247	6	762 bytes	6	100.00%	3	483 bytes	3	279 bytes	1
163.70.144.60	8	716 bytes	8	100.00%	4	360 bytes	4	356 bytes	1
163.181.214.12	18	10 kB	18	100.00%	10	7 kB	8	3 kB	1
104.18.39.21	22	2 kB	22	100.00%	12	1 kB	10	956 bytes	1
139.45.197.106	33	18 kB	33	100.00%	16	10 kB	17	8 kB	1
104.16.79.73	30	19 kB	30	100.00%	18	15 kB	12	3 kB	1
142.250.146.95	31	13 kB	48	64.58%	18	9 kB	13	4 kB	1
172.217.119.4	36	13 kB	36	100.00%	20	8 kB	16	5 kB	1
172.64.155.209	57	31 kB	57	100.00%	30	6 kB	27	25 kB	1
23.52.73.105	109	100 kB	109	100.00%	76	97 kB	33	3 kB	1
192.168.0.102	363	208 kB	2,307	15.73%	148	53 kB	215	155 kB	1

Protocol	Percent Packets
Frame	100.0
Ethernet	100.0
Internet Protocol Version 4	100.0
Transmission Control Protocol	100.0
Transport Layer Security	33.9
Data	0.3

Most Active Source IP – 192.168.0.102

Most Active Destination IP - 192.168.0.102

Number/type of communications – TCP and TLS

2. DNS Investigation

Source IP: 192.168.0.102 (The client workstation/device under investigation).

Destination IP: 192.168.0.1 (The local DNS recursive resolver/default gateway router).

Protocol: DNS (Standard unencrypted UDP port 53 traffic).

Query Types: Standard Type A (IPv4 address lookup) and Type HTTPS (used by modern browsers like Chrome/Firefox to look up service binding parameters and optimize connection security before launching HTTP/2 or HTTP/3 sessions).

3. TCP Investigation

tcp.flags.syn == 1						
No.	Time	Source	Destination	Protocol	Length	Info
585	27.991242900	104.16.79.73	192.168.0.102	TCP	66	443 → 34939 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 SACK_PERM WS=8192
2159	30.748884300	139.45.197.106	192.168.0.102	TCP	58	443 → 53406 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1440
466	27.900512100	142.250.146.95	192.168.0.102	TCP	66	443 → 18073 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
1926	28.765483300	163.181.214.12	192.168.0.102	TCP	66	443 → 41939 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1440 SACK_PERM WS=512
1440	28.512945300	172.217.119.4	192.168.0.102	TCP	66	443 → 12434 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
462	27.892618500	192.168.0.102	142.250.146.95	TCP	66	18073 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
480	27.922858700	192.168.0.102	104.16.79.73	TCP	66	34939 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1422	28.507030900	192.168.0.102	172.217.119.4	TCP	66	12434 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
1698	28.638580300	192.168.0.102	163.181.214.12	TCP	66	41939 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
2156	30.613925700	192.168.0.102	139.45.197.106	TCP	66	53406 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
2200	31.092299700	192.168.0.102	23.52.73.105	TCP	66	53407 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
2201	31.135120400	23.52.73.105	192.168.0.102	TCP	66	443 → 53407 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1384 SACK_PERM WS=128
2202	31.135121500	23.52.73.105	192.168.0.102	TCP	66	[TCP Retransmission] 443 → 53407 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1384 SACK_PERM WS=128

The local host (192.168.0.102) sends [SYN] packets from various high-numbered source ports (e.g., 18073, 34939, 12434) to port 443 on external IP addresses like 142.250.146.95 (Google) and 104.16.79.73 (Cloudflare)

tcp.flags.syn == 1 && tcp.flags.ack == 1						
No.	Time	Source	Destination	Protocol	Length	Info
585	27.991242900	104.16.79.73	192.168.0.102	TCP	66	443 → 34939 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 SACK_PERM WS=8192
2159	30.748884300	139.45.197.106	192.168.0.102	TCP	58	443 → 53406 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1440
466	27.900512100	142.250.146.95	192.168.0.102	TCP	66	443 → 18073 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
1926	28.765483300	163.181.214.12	192.168.0.102	TCP	66	443 → 41939 [SYN, ACK] Seq=0 Ack=1 Win=42340 Len=0 MSS=1440 SACK_PERM WS=512
1440	28.512945300	172.217.119.4	192.168.0.102	TCP	66	443 → 12434 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK_PERM WS=256
2201	31.135120400	23.52.73.105	192.168.0.102	TCP	66	443 → 53407 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1384 SACK_PERM WS=128
2202	31.135121500	23.52.73.105	192.168.0.102	TCP	66	[TCP Retransmission] 443 → 53407 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1384 SACK_PERM WS=128

It shows that remote servers (like 142.250.146.95 and 104.16.79.73) have actively received a connection request from the host machine (192.168.0.102) and are agreeing to open a communication channel.

tcp.flags.reset == 1						
No.	Time	Source	Destination	Protocol	Length	Info
348	27.742846800	150.171.27.10	192.168.0.102	TCP	54	443 → 44848 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2346	35.440446600	150.171.27.10	192.168.0.102	TCP	54	443 → 56675 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
2313	32.421184400	150.171.27.12	192.168.0.102	TCP	54	443 → 8235 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

RST packet instantly kills the connection. This usually happens because a packet arrived for a closed socket, a firewall blocked the traffic, or the server abruptly refused the handshake request.

4. Suspicious Activity

No.	Time	Source	Destination	Protocol	Length	Info
308	27.616364300	192.168.0.1	192.168.0.102	DNS	138	Standard query response 0x053a HTTPS ogads-pa.clients6.google.com SOA ns1.google.com
309	27.616369200	192.168.0.1	192.168.0.102	DNS	268	Standard query response 0xe2c9 A ogads-pa.clients6.google.com A 142.250.195.42 NS ns3.google.com NS ns4.google.com NS ns1.google.com
355	27.793770400	192.168.0.1	192.168.0.102	DNS	489	Standard query response 0x3c4a A cdnjs.cloudflare.com A 104.17.24.14 A 104.17.25.14 NS l.root-servers.net NS c.root-servers.net NS e.root-servers.net NS f.root-servers.net NS g.root-servers.net NS h.root-servers.net NS i.root-servers.net NS j.root-servers.net NS k.root-servers.net NS l.root-servers.net NS m.root-servers.net NS n.root-servers.net NS o.root-servers.net NS p.root-servers.net NS q.root-servers.net NS r.root-servers.net NS s.root-servers.net NS t.root-servers.net NS u.root-servers.net NS v.root-servers.net NS w.root-servers.net NS x.root-servers.net NS y.root-servers.net NS z.root-servers.net
356	27.793773500	192.168.0.1	192.168.0.102	DNS	530	Standard query response 0xace2 HTTPS cdnjs.cloudflare.com HTTPS NS b.root-servers.net NS e.root-servers.net NS c.root-servers.net NS f.root-servers.net NS g.root-servers.net NS h.root-servers.net NS i.root-servers.net NS j.root-servers.net NS k.root-servers.net NS l.root-servers.net NS m.root-servers.net NS n.root-servers.net NS o.root-servers.net NS p.root-servers.net NS q.root-servers.net NS r.root-servers.net NS s.root-servers.net NS t.root-servers.net NS u.root-servers.net NS v.root-servers.net NS w.root-servers.net NS x.root-servers.net NS y.root-servers.net NS z.root-servers.net
389	27.834104600	192.168.0.1	192.168.0.102	DNS	143	Standard query response 0x6f85 HTTPS encrypted-tbn0.gstatic.com SOA ns1.google.com
390	27.834123300	192.168.0.1	192.168.0.102	DNS	531	Standard query response 0x9896 A encrypted-tbn0.gstatic.com A 142.250.146.139 A 142.250.146.102 A 142.250.146.100 A 142.250.146.101
457	27.891099800	192.168.0.1	192.168.0.102	DNS	137	Standard query response 0x0e3c HTTPS fonts.googleapis.com SOA ns1.google.com
458	27.891101400	192.168.0.1	192.168.0.102	DNS	473	Standard query response 0x4c35 A fonts.googleapis.com A 142.250.146.95 NS k.root-servers.net NS b.root-servers.net NS f.root-servers.net NS g.root-servers.net NS h.root-servers.net NS i.root-servers.net NS j.root-servers.net NS l.root-servers.net NS m.root-servers.net NS n.root-servers.net NS o.root-servers.net NS p.root-servers.net NS q.root-servers.net NS r.root-servers.net NS s.root-servers.net NS t.root-servers.net NS u.root-servers.net NS v.root-servers.net NS w.root-servers.net NS x.root-servers.net NS y.root-servers.net NS z.root-servers.net
477	27.921560700	192.168.0.1	192.168.0.102	DNS	536	Standard query response 0x7951 HTTPS static.cloudflareinsights.com HTTPS NS a.root-servers.net NS e.root-servers.net NS c.root-servers.net NS f.root-servers.net NS g.root-servers.net NS h.root-servers.net NS i.root-servers.net NS j.root-servers.net NS k.root-servers.net NS l.root-servers.net NS m.root-servers.net NS n.root-servers.net NS o.root-servers.net NS p.root-servers.net NS q.root-servers.net NS r.root-servers.net NS s.root-servers.net NS t.root-servers.net NS u.root-servers.net NS v.root-servers.net NS w.root-servers.net NS x.root-servers.net NS y.root-servers.net NS z.root-servers.net
478	27.921569000	192.168.0.1	192.168.0.102	DNS	498	Standard query response 0x3b65 A static.cloudflareinsights.com A 104.16.79.73 A 104.16.80.73 NS f.root-servers.net NS a.root-servers.net NS b.root-servers.net NS c.root-servers.net NS d.root-servers.net NS e.root-servers.net NS g.root-servers.net NS h.root-servers.net NS i.root-servers.net NS j.root-servers.net NS k.root-servers.net NS l.root-servers.net NS m.root-servers.net NS n.root-servers.net NS o.root-servers.net NS p.root-servers.net NS q.root-servers.net NS r.root-servers.net NS s.root-servers.net NS t.root-servers.net NS u.root-servers.net NS v.root-servers.net NS w.root-servers.net NS x.root-servers.net NS y.root-servers.net NS z.root-servers.net
727	28.122019800	192.168.0.1	192.168.0.102	DNS	484	Standard query response 0x95e2 A nekostream.site A 172.67.189.251 A 104.21.81.144 NS g.root-servers.net NS f.root-servers.net NS e.root-servers.net NS d.root-servers.net NS c.root-servers.net NS b.root-servers.net NS a.root-servers.net
728	28.128270400	192.168.0.1	192.168.0.102	DNS	544	Standard query response 0x9be3 HTTPS nekostream.site HTTPS NS i.root-servers.net NS h.root-servers.net NS l.root-servers.net NS k.root-servers.net NS j.root-servers.net NS m.root-servers.net NS n.root-servers.net NS o.root-servers.net NS p.root-servers.net NS q.root-servers.net NS r.root-servers.net NS s.root-servers.net NS t.root-servers.net NS u.root-servers.net NS v.root-servers.net NS w.root-servers.net NS x.root-servers.net NS y.root-servers.net NS z.root-servers.net
753	28.201153700	192.168.0.1	192.168.0.102	DNS	125	Standard query response 0x1204 HTTPS play.google.com SOA ns1.google.com
756	28.202193800	192.168.0.1	192.168.0.102	DNS	255	Standard query response 0x1f41 A play.google.com A 172.217.160.110 NS ns2.google.com NS ns4.google.com NS ns1.google.com NS ns3.google.com
1407	28.501690700	192.168.0.1	192.168.0.102	DNS	479	Standard query response 0x5c8 HTTPS fonts.gstatic.com HTTPS NS h.root-servers.net NS f.root-servers.net NS a.root-servers.net NS e.root-servers.net NS c.root-servers.net NS g.root-servers.net NS i.root-servers.net NS j.root-servers.net NS k.root-servers.net NS l.root-servers.net NS m.root-servers.net NS n.root-servers.net NS o.root-servers.net NS p.root-servers.net NS q.root-servers.net NS r.root-servers.net NS s.root-servers.net NS t.root-servers.net NS u.root-servers.net NS v.root-servers.net NS w.root-servers.net NS x.root-servers.net NS y.root-servers.net NS z.root-servers.net
1408	28.501692800	192.168.0.1	192.168.0.102	DNS	470	Standard query response 0xf4d0 A fonts.gstatic.com A 192.178.193.94 NS h.root-servers.net NS e.root-servers.net NS i.root-servers.net NS j.root-servers.net NS k.root-servers.net NS l.root-servers.net NS m.root-servers.net NS n.root-servers.net NS o.root-servers.net NS p.root-servers.net NS q.root-servers.net NS r.root-servers.net NS s.root-servers.net NS t.root-servers.net NS u.root-servers.net NS v.root-servers.net NS w.root-servers.net NS x.root-servers.net NS y.root-servers.net NS z.root-servers.net
1410	28.502632000	192.168.0.1	192.168.0.102	DNS	148	Standard query response 0x0d67 HTTPS content-autofill.googleapis.com SOA ns1.google.com
1411	28.502783700	192.168.0.1	192.168.0.102	DNS	540	Standard query response 0x25a7 A content-autofill.googleapis.com A 172.217.119.4 A 172.217.117.4 A 172.217.112.4 A 172.217.118.4
1541	28.560164000	192.168.0.1	192.168.0.102	DNS	149	Standard query response 0x56b3 HTTPS dump.mewcdn.online SOA darling.cs.atrustdns.com

dump.mewcdn.online (Packet 1541) This domain stands out the most. The query features an unusual structure, and the root authority points to atrustdns.com.

nekostream.site (Packet 728) nekostream.site is an infrastructure layer typically associated with unauthorized online video streaming, manga repositories, or media mirroring platforms.

Time	Source IP	Destination IP	Port	Protocol	Observation	Security Significance
27.616	192.168.0.1	192.168.0.102	53	DNS	DNS response for ogads-pa.clients6.google.com	Normal DNS response; shows domain resolution activity.
28.561	192.168.0.1	192.168.0.102	53	DNS	DNS response involving dump.mecdnn.online	Unexpected domain activity may warrant further investigation.
28.512	192.168.0.102	172.217.119.4	443	TCP	SYN packet sent to destination port 443	Indicates an attempted HTTPS TCP connection.
27.742	150.171.27.10	192.168.0.102	443	TCP	RST/ACK received	Connection was reset; may occur for normal connection/availability reasons.
32.421	150.171.27.12	192.168.0.102	443	TCP	RST/ACK received	Connection reset by the remote endpoint.

TASK 12 – Vulnerability Assessment

Finding	Evidence	Risk	Potential Impact	Recommendation
vsftpd 2.3.4 is outdated and has a known security issue	Nmap -sV identified FTP – vsftpd 2.3.4 on port 21	Critical	The version is associated with a known backdoor vulnerability that can potentially allow unauthorized remote access.	Remove/disable FTP if unnecessary.

Telnet service is exposed	Nmap identified Telnet on TCP port 23	High	Telnet transmits authentication information without adequate encryption, allowing credentials and session data to potentially be exposed through network interception.	Disable Telnet and use SSH for remote administration.
Outdated Apache HTTP server	Nmap -sV identified Apache 2.2.8 on port 80	High	An outdated web server may contain publicly known vulnerabilities that could allow unauthorized access or compromise of the web service.	Upgrade Apache to a currently supported version and regularly apply security patches.
Metasploitable root shell service exposed	Nmap identified TCP port 1524 associated with the Metasploitable root shell service	Critical	An exposed root-level service can provide extremely high-privilege unauthorized access if reachable by an attacker.	Disable the service and restrict unnecessary ports with firewall rules.
Large number of network services exposed	Nmap -sV found 22 open TCP ports on 192.168.27.129	High	Multiple exposed services increase the attack surface and provide more opportunities for exploitation or unauthorized access.	Disable unnecessary services and use firewall/ACL rules to restrict access to required services only.

TASK 13 – Security Hardening & Before/After Testing

1. Before Hardening

nmap -sV 192.168.27.129

```
(nehawalmiki@kali)-[~/Desktop]
$ nmap -sV 192.168.27.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-21 08:22 -0500
Nmap scan report for 192.168.27.129
Host is up (0.0011s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
```

Finding	Before
FTP	Port 21 open
HTTP	Port 80 open
Telnet	Port 23 open

2. Apply 3 Security Improvements

Disable FTP - sudo service vsftpd stop

Disable telnet - sudo service telnetd stop

Security Issue	Before	Action Taken	After	Improvement
FTP service exposed	21/tcp – open – FTP	Disabled the FTP service	21/tcp – closed	Removed unnecessary FTP exposure and reduced the attack surface
Telnet service exposed	23/tcp – open – Telnet	Disabled Telnet service	23/tcp – closed	Removed an unnecessary remote-access service

🔴 TASK 14 – Final Mini Security Assessment

1. Objective

The objective of this assessment was to perform a basic security assessment of an authorized lab network using network discovery, Nmap scanning, Wireshark traffic analysis, vulnerability assessment, and security hardening.

2. Network Discovery

Devices Identified

The lab environment consisted of:

Device	Role	IP Address
Kali Linux	Security testing/scanning machine	192.168.27.128
Metasploitable2	Target/lab server	192.168.27.129

Services Exposed

The Nmap -sV scan of Metasploitable2 identified multiple exposed services. Your actual scan showed 23 open TCP ports. Some important examples from your screenshot were:

Port Service Version/Information

21	FTP	vsftpd 2.3.4
22	SSH	OpenSSH
23	Telnet	Linux telnetd
25	SMTP	SMTP service

3. Traffic Analysis

Wireshark was used to capture and analyze traffic generated within the authorized lab environment.

Protocols Observed

Your capture showed protocol layers including:

- Ethernet
- IPv4
- TCP
- TLS
- Data

4. Security Assessment

Based on the Nmap results and vulnerability assessment, several security findings were identified.

Finding	Evidence	Risk
Multiple unnecessary/exposed services	Nmap identified 23 open TCP ports	High
Telnet exposed	Port 23/TCP open	High
FTP exposed	Port 21/TCP open with FTP service	Medium/High
Legacy remote-service exposure	Port 512/TCP rexecd detected	High
Multiple legacy services/software	Nmap service/version detection	High

5. Hardening

Security hardening was performed on the authorized Metasploitable2 lab machine.

The objective was to reduce unnecessary network exposure by disabling services that were not required for the lab.

Before and After

Security Issue	Before	Action Taken	After	Improvement
FTP exposed	Port 21 open	FTP service disabled	<actual result>	Reduced FTP exposure
Telnet exposed	Port 23 open	Telnet disabled	<actual result>	Removed unnecessary remote-access service

Security Issue	Before	Action Taken	After	Improvement
rexecd exposed	Port 512 open	rexecd disabled	<actual result>	Reduced legacy remote-service exposure

Did the Attack Surface Reduce?

Yes. The attack surface was reduced because the number of exposed services decreased after hardening. The before-and-after Nmap comparison provided evidence that the selected services were no longer accessible on their respective ports.

6. Final Recommendations

1. Disable unnecessary services
 2. Replace insecure protocols
 3. Apply regular security updates
 4. Implement firewall restrictions
 5. Continuously monitor network traffic and services
-